

Online Resource 1: Rotational constraints on nuclear population memory in a cryptochrome reaction model

Hikaru Wakaura^{1*}

^{1*}QIRI (Quantum Integrated Research Institute Inc.), Tokyo, 107-0061, Japan.

Corresponding author(s). E-mail(s): h.wakaura@qiri.co.jp;

Abstract

Supplementary definitions, model provenance, numerical checks and limits of the motion-dependent revision. Numerical comparisons below are conditional computations, not measured neuronal parameters.

S1 Inputs, chemical states and cycle

The reference archive is `data/population-memory/latest_case.npz`; its SHA-256 is recorded in Online Resource 2. It is unchanged by the rotation, binding and electric-response comparisons. H denotes anionic closed-shell HQ, R closed-shell oxidised flavin and E neutral open-shell semiquinone. N5 and N10 are both spin-1 ¹⁴N. A second electron is included in RP. The dimensions of H/R, E and RP are 9, 18 and 36, respectively. The reduced system excludes other nuclei and additional oxygen-electron relaxation. Electron exchange and electron dipolar coupling are zero. These omissions can affect radical-pair yields and should not be read as statements about their absence in a protein.

The Hamiltonians are expressed in angular-frequency units, with dimensionless spin operators. The nuclear contribution is

$$H_{n,s} = -\gamma_n \mathbf{B} \cdot \sum_{k=1}^2 \mathbf{I}_k + \frac{eQ}{2\hbar} \sum_{k=1}^2 \sum_{a,b} V_{ab}^{(s,k)} \frac{I_{k,a} I_{k,b} + I_{k,b} I_{k,a}}{2}, \quad (\text{S1})$$

for $I = 1$, where $Q = 2.04430 \times 10^{-2}$ barn is the adopted quadrupole-moment input and $\gamma_n = (4.03761 \times 10^{-1})\mu_N/\hbar$. These are declared reference constants, not fitted relaxation quantities. E adds the electron Zeeman term and $\sum_{k,a,b} A_{ab}^{(k)} S_{1,a} I_{k,b}$. RP adds a second electron Zeeman term; the second electron has no hyperfine coupling in this reduced model. EFG is in V m^{-2} and HFC is converted to rad s^{-1} . The tensors used by the calculation are preserved through the full Hamiltonian arrays in the frozen archive.

For the local bath, let $x_a(t)$ be coefficients of the symmetric traceless EFG/HFC tensor operators and $\delta x_a = x_a - \bar{x}_a$. The retained coefficient matrix is the sliding-window estimate

$$K_{ab} = \frac{1}{T} \left\langle \left(\int_t^{t+T} \delta x_a(u) du \right) \left(\int_t^{t+T} \delta x_b(u) du \right) \right\rangle_t, \quad (\text{S2})$$

with $T = 1.00000$ ps, a 6.00000 ps trajectory and 2.00000 fs sampling. Diagonalising this positive-semidefinite matrix defines the Lindblad operators and nonnegative weights. This is a finite-window zero-frequency spectral estimate, not an independently measured infinite-time correlation function. The state-specific mean tensors and the corresponding transferred local-noise construction are frozen between the motion comparisons. Original data paths and coefficient-building source are listed in Online Resource 2.

Write $\mathcal{W}_s(r) = r(rI - \mathcal{G}_s)^{-1}$ for an exponential wait in chemical state s . This follows from $\int_0^\infty r e^{-rt} e^{\mathcal{G}_s t} dt$. Let $\mathcal{J}$ inject the triplet-mixture electron state, $\mathcal{L}_{\text{RP}}$ contain coherent evolution, motion, local noise and reaction loss, and $\mathcal{P}$, $\mathcal{E}$ be integrated product and escape maps:

$$\mathcal{P}\rho = k_S \langle S | (-\mathcal{L}_{\text{RP}})^{-1} \mathcal{J}\rho | S \rangle, \quad (\text{S3})$$

$$\mathcal{E}\rho = k_E \text{Tr}_{e_2} [(-\mathcal{L}_{\text{RP}})^{-1} \mathcal{J}\rho]. \quad (\text{S4})$$

The loss term is $-\{k_S P_S + k_E I, \rho\}/2$ [1]. This specifies the chosen reaction super-operator; alternatives may add reactive exchange or dephasing [2]. The complete pre-boundary cycle is

$$\mathcal{M} = \mathcal{W}_H(10) \mathcal{W}_R(10) \left[\mathcal{W}_R(10) \mathcal{P} + \text{Tr}_e \mathcal{W}_E(4) \mathcal{E} \right]. \quad (\text{S5})$$

Rates in this equation have units s^{-1} . There are two R waits on the product route: product recovery followed by reduction. On the escape route, electron tracing follows the E wait and precedes R reduction. $k_S = 10^7 \text{ s}^{-1}$ and $k_E = 4 \times 10^6 \text{ s}^{-1}$. All-waits acceleration changes the rates of these waits while leaving k_S, k_E fixed. The H-only protection control suppresses rigid-body rotational diffusion in H alone; all-waits protection suppresses it in H, R and E while retaining the frozen local bath. RP still rotates.

The initial nuclear state is $I_9/9$. The freely rotating ensemble initially is isotropic; binding runs initialise the specified motion-only equilibrium. At each completed cycle

the chosen full, population or reset rule is applied. The next product yield after cycle 20 defines the reported observable. The nuclear reset retains each orientation/binding marginal, and does not remove the current reaction’s quantum dynamics. No fixed elapsed-time preparation is implied.

Table S1 Input status and limits.

Input	Origin	Limit
Static H/R/E interactions	Computed fixed embedded geometries	Finite fragment, state and basis choices
Local noise	6 ps oxidised-state trajectory, 1 ps window, 298 K	Transferred between states; long-time Markovian extrapolation
Reaction rates and preparation	Prescribed scenario	No neuronal rate fit or native preparation assay
Rotational and binding conditions	Specified conditional sweep	No inferred brain distribution or binding partner
Electric dipole and field	Assumed effective coupling	No measured local field, screening or membrane depth
Electronic susceptibility	Matched fixed-geometry response	No field-relaxed geometry, MM polarisation or new bath
Circuit conversion	Prescribed source, gain and network	No chemical-to-channel or synaptic calibration

The reference local bath is represented in positive-coefficient Lindblad form [3]. Relaxation from fluctuations has precedents in cryptochrome spin dynamics [4], but that literature does not determine this closed-shell nuclear bath. The calculations concern the retained nuclei in one reduced reaction model, not a validated model of a particular brain region.

S2 Free motion, spectra and counterfactual limits

In the body frame the field direction is $\mathbf{n} = R^{-1}\hat{\mathbf{z}}$. Angular functions are expanded in spherical harmonics through degree L . The covariant generator couples orbital rotation and spin-frame rotation:

$$\mathcal{G}_s = -i \text{ad } H_{s,\text{body}} + \mathcal{D}_{s,\text{local}} - iB \sum_a n_a \text{ad } Z_{s,a} - \frac{1}{6\tau_2} \sum_a (L_{\text{orb},a} + \text{ad } J_{s,a})^2. \quad (\text{S6})$$

The total J_s includes both nuclear spins and the electrons present in s . A common molecular orientation persists across chemical transitions. Orientations are not independently reinitialised at every reaction. The finite-field terms and rotational operators are projected onto the angular basis. Waiting resolvents are solved with explicit identity/traceless decomposition to avoid stiff-solver contamination of tiny spin differences. Trace, Hermiticity and sampled-density positivity checks are retained.

The spectral quantity reported in the main article is $1/\min_{\lambda \neq 0}[-\Re \lambda]$ for decaying nontrivial H modes, with numerical selection and independent eigenvalue checks in the archived scripts. It need not equal a measured longitudinal T_1 : a prepared distribution can have negligible overlap with the slowest mode, and an observable can be insensitive to that mode. The fixed-orientation 1.63474 s half-decay refers instead to decay of a particular prepared reaction-readable contrast during isolated H storage. The all-waits-protected 1.53143 s half-decay uses another prepared state and readout. Neither is a binding-sweep lifetime.

Table S2 Extreme all-waits acceleration illustrates the boundary of the population approximation. The multiplier applies to H/R/E wait rates only. These are not fitted physiological rates.

Multiplier	Δq	$q_{\text{full}} - q_{\text{pop}}$	Mean preparation (μs)
10^4	2.88347×10^{-9}	6.45847×10^{-10}	800.83806
10^6	1.41208×10^{-5}	6.49630×10^{-6}	11.26778
10^8	2.00984×10^{-4}	2.80920×10^{-5}	3.37123

For the 10^8 multiplier, adding 100.00000 μs of freely rotating H storage reduces the contrast to approximately 7.40398×10^{-10} . At 1.00000 ms it falls far below the reporting resolution; the machine-scale value is not assigned physical significance. Accelerated writing therefore does not demonstrate long free storage. A reset-prepared single-reaction field response is separately retained: changing B from zero to 50.00000 μT gives a yield difference of 7.92616×10^{-4} in that probe protocol. It is not the full-reset history contrast.

Free-cycle comparisons use $L = 3$. Independent $L = 4$ absolute-yield differences are 6.12665×10^{-12} , 6.12721×10^{-10} and 1.10235×10^{-6} for the three increasing τ_2 values. The slowest-rotation absolute yield was additionally assessed at higher angular cutoffs in the original investigation. The 10^{-10} history reporting resolution must not be substituted for a bound on absolute yield error. Direct traceless-state calculations support the unresolved history result without relying only on subtracting two nearly equal yields.

S3 Binding, bound wobble and the meaning of population

The binding model carries mobile angular coefficients and bound spin density at anchor quadrature nodes. Capture and release conserve spin, occur in all chemical states and retain the orientation information of the adopted coarse graining. For isotropic capture, the input fraction satisfies $f = k_{\text{on}}/(k_{\text{on}} + k_{\text{off}})$ in the motion-only problem. Additional controls use $k_{\text{on}}(\mathbf{n}) = k_{\text{on}}[1 + aP_2(n_z)]$ with $a = -0.90000, 0.00000, 1.80000$. This changes which orientations are captured. The reacting ensemble can have a different bound fraction because reaction and spin dynamics weight the motion sectors differently.

The bound angular covariance is $\langle \theta_a(t) \theta_b(0) \rangle = \delta_{ab} (\theta_{\text{rms}}^2 / 3) e^{-|t|/\tau_w}$ at the level of the assumed small-angle noise. Expanding the Hamiltonian to linear order gives Hermitian derivative operators $V_a = -i[J_a, H_{\text{body}}]$. The fast-noise limit yields the dissipator in main-text equation (4). A six-node positive rotation cubature averages the Hamiltonian and frozen local dissipator. This representation includes mean-interaction changes as well as relaxation, so arbitrarily small added wobble need not monotonically reduce every yield contrast. It is a reduced Gaussian-noise model, not a pathwise OU solution.

Full and population propagation share all within-cycle physics. In a pure state, populations are $|c_j|^2$, while c_j are amplitudes and off-diagonal density-matrix entries encode relative-phase information. In a mixed state the same distinction uses ρ_{jj} and ρ_{jk} . The H-boundary dephasing basis depends on orientation and, for bound molecules, the averaged well Hamiltonian. A joint diagonal distribution need not factorise into two single-nucleus distributions. Thus, removing inter-cycle coherence preserves classical correlations as well as unequal populations. Wong et al. [5], equation (8), already propagate joint diagonal probabilities in a different basis and photocycle. Their phrase “full retention” refers to reaction-branch retention, not to our full density-matrix rule.

A nuclear-history and motion-marginal decomposition is useful because repeated reset changes the prepared ensemble indirectly. Let x_F, x_R be the final full and repeatedly reset spin-motion states, $\mathcal{R}$ reset only the nuclear factor at each motion coordinate, and ℓ be the next-yield functional. Then

$$\Delta q_{\text{spin}} = \ell(x_F) - \ell(\mathcal{R}x_F), \quad (\text{S7})$$

$$\Delta q_{\text{motion}} = \ell(\mathcal{R}x_F) - \ell(x_R), \quad (\text{S8})$$

$$\Delta q = \Delta q_{\text{spin}} + \Delta q_{\text{motion}}. \quad (\text{S9})$$

This is an intervention-defined decomposition, not a unique decomposition of all possible spin-motion correlations. The representative 1.00000° , 1.00000 s, $f = 0.50000$ case has $\Delta q_{\text{spin}} = 1.29817 \times 10^{-3}$ and $\Delta q_{\text{motion}} = -9.55628 \times 10^{-6}$.

There are 58 unique molecular binding conditions, 15 additional cutoff runs and three additional quadrature runs. Selected $L = 2/3$ maximum differences are 2.61946×10^{-6} for individual yields and 3.06281×10^{-7} for the contrast. Increasing selected quadrature grids from 55 to 171 nodes changes Δq by at most 5.19316×10^{-9} . Independent H resolvent, reaction adjoint and static-cycle power checks address separate numerical operations. A finite-state telegraph comparison to the reduced H wait map gives relative differences 1.12115×10^{-8} , 7.16336×10^{-5} and 5.98643×10^{-3} at RMS angles 0.10000 , 1.00000 and 5.00000° with $\tau_w = 10.00000$ ps. Telegraph noise is not OU noise, and this is not a full-cycle physical-error estimate. The larger-angle results remain conditional on the reduction.

S4 Electric orientation and matched electronic response

The electric orientation calculation adds the potential $-\kappa \cos \theta$ to the free rotational problem and applies the corresponding equilibrium weighting and detailed-balance

capture. $\kappa = \mu E / (k_B T)$ uses an assumed effective dipole 1000 D and $T = 310$ K, with tested values 0.00000, 0.10911, 1.09109, 10.91094. The microscopic local field is not inferred from a measured membrane voltage. Membrane-peptide reorientation [6], Abstract, is a physical precedent for torque, not evidence for these cryp-tochrome inputs. Spin-1 quadrupolar sensitivity to EFG [7], Introduction, is a separate interaction and does not demonstrate membrane-induced spin polarisation.

The reference tensors are matched as follows. H and R reload the embedded radius-12 Å checkpoints with def2-SVPD. E reloads the neutral-SQ full minimum-image environment with uncontracted pcJ-2 on nitrogen. Frozen capped flavin geometries and static MM charges are retained. Electronic response uses PySCF 2.13.0 with the properties package 0.1.0, B3LYP, grid 4 and density fitting with def2-universal-jkfit. PySCF’s general electronic-structure framework is described in [8]; this citation does not validate the present geometry or response approximations.

Define $\chi_{aij}^V = \partial V_{ij} / \partial E_a$ and $\chi_{aij}^A = \partial A_{ij} / \partial E_a$. All three field directions are computed analytically. The susceptibilities transform as third-rank tensors through the same local rotations as the baseline tensors. Static external MM EFG is not added a second time. A spatially uniform field contributes no direct EFG, but changes the electronic density and hence the nuclear EFG. H/R are closed shell and have no unpaired-electron HFC in this nonrelativistic model. E includes Fermi-contact and spin-dipolar HFC. The changed $H(E)$ enters both common rotation and the bound-wobble derivatives. The original local-bath coefficients remain fixed.

Table S3 Maximum directional linear response at 14.00000 MV/m. EFG uses the Frobenius norm divided by the corresponding zero-field QM EFG norm. Different maxima can occur at different field directions.

State	Nucleus	EFG change (%)	Isotropic HFC change (MHz)	Baseline (MHz)
H	N5	0.02746	—	—
H	N10	0.02199	—	—
R	N5	0.02418	—	—
R	N10	0.04879	—	—
E	N5	0.04085	0.01143	11.03879
E	N10	0.04549	0.00887	6.12097

Zero-field tensors were recovered from the matched electronic checkpoints. Central finite-field SCF pairs in direction $(1, 2, 3) / \sqrt{14}$ use ± 14.00000 MV/m for H/R/E and ± 7.00000 MV/m for H/E, five pairs in total. Relative derivative discrepancies (EFG; HFC where applicable) are H: 0.00877% at 14 MV/m and 0.00812% at 7 MV/m; R: 0.00992% at 14 MV/m; E: 0.03617%/0.14651% at 14 MV/m and 0.14934%/0.14024% at 7 MV/m. Finite-field checks do not independently test every tensor direction. Even-in-field residuals combine nonlinearity and SCF residuals; no small quadratic coefficient is inferred from them.

S5 Deformation of the conditional binding well

For each anchor direction, the rotational well is $U/k_B T = c|\log R|^2/2 - \kappa \mathbf{n} \cdot R\mathbf{p}$, with $c = 3/\theta_{\text{rms},0}^2$. The nonlinear minimum is found before differentiating in the tangent space of $\text{SO}(3)$. The Hessian K gives the Gaussian covariance, and fixed mobility gives $\tau_i = c\tau_0/\lambda_i$. Derivative operators are Hermitian projected before finite-angle differentiation to prevent roundoff amplification from generating an artificial nonunit term. The effect of this Hermitian projection on the calculated yields is at most 2.37588×10^{-14} .

For uniformly distributed anchors A , write the absolute dipole orientation as $AR\mathbf{p}$. The joint orientation partition factor contains

$$\int dR e^{-c|\log R|^2/2} \int dA e^{\kappa \hat{\mathbf{E}} \cdot AR\mathbf{p}}. \quad (\text{S10})$$

The inner integral is independent of R by Haar invariance and equals the corresponding free-dipole factor. With the same dipole in the free and bound sectors, this factor cancels in their equilibrium ratio. Consequently a total bound-fraction increase is absent under these symmetry assumptions. Field-induced differences between free/bound dipoles, anisotropic anchor priors or translational binding energies would invalidate this cancellation and require additional inputs.

The local Laplace approximation has a normalisation error of -3.59524×10^{-5} at 5.00000° , $\kappa = 10.91094$; it is normalised away. Prescribed $f = 0.50000$ and $k_{\text{off}} = 1.00000 \text{ s}^{-1}$ are maintained. Capture weights change by anchor direction, but a new native k_{off} is not calculated. A dissociation barrier and binding partner have not been specified.

Fifty well conditions include zero, weak and strong fields, sign controls and anchor angles 0, 45, 90, 135, 180 degrees. At 5.00000° and strong field, the parallel-well correlation times span 9.73049–10.00000 ps; the antiparallel range is 10.00000–10.28486 ps. The perpendicular mean shift is 1.58632° . Independent differentiation verifies the Hessian. Nonlinear three-dimensional Gauss–Hermite integration including Haar measure differs from harmonic covariance by at most 0.04736% in the selected comparisons; the order-7/11 difference is at most 1.84447×10^{-9} . These verify a specified mathematical restraint, not its correspondence to native binding.

The four-arm yield table in the main article uses $L = 2$ throughout. Combined arms also use $L = 3$; the maximum cutoff difference is 1.32617×10^{-6} for a yield and 4.39672×10^{-10} for a contrast. Independent direct and iterative H solves differ by 3.33145×10^{-12} , and an independent third-rank transformation check differs by 4.44089×10^{-16} . Maximum trace and sampled Hermiticity errors are 5.02963×10^{-14} and 2.86167×10^{-12} ; the minimum sampled density eigenvalue is 0.05097. These sampled checks do not prove positivity over every orientation. The field-coupling dataset comprises three electronic states, five finite-field pairs, ten bound-cycle conditions, one free-cycle condition and nine H spectra.

The original 298 K bath is not regenerated at the 310 K used to convert dipole energy to κ . Missing mechanisms include field-dependent local noise, protonation and geometry changes, MM polarisation, solvent/side-chain dynamics, reaction and

proton-transfer rates, spatial gradients, spin-orbit/ g response and membrane-depth coupling. The three mechanisms evaluated here do not exhaust possible electric-field effects.

S6 Conditional stochastic circuit

The molecular input is $q_0 = q_{\text{reset}}$ and $q_1 = q_{\text{full}}$ for each selected zero-electric-field binding condition. A pool of $N = 10000$ independent binary units is stationary at q_0 before a common preparation at cue time $t_c = 200.00000$ ms. That preparation draws the two coupled binomial pools with probabilities q_0, q_1 . Subsequent exact finite-population transitions relax to the same q_0 with survival $e^{-\Delta t/\tau}$, $\tau = 200.00000$ ms and $\Delta t = 10.00000$ ms. Common random numbers reduce Monte Carlo noise but do not add biological samples.

For pool fraction x , release probabilities are

$$p_i(x) = p_0 + \alpha_i C \tanh\left[\frac{g(x - q_0)}{C}\right], \quad C = 0.95000 \min(p_0, 1 - p_0)k. \quad (\text{S11})$$

Here $p_0 = 0.45000$, $g = 1024$, $k = 8$ and $\alpha_i = 1/8$ on selected connections, zero elsewhere. The architecture has 64 connections. A fixed seed (26091681) generates weights, and the eight largest $w_i e^{-d_i/(40.00000 \text{ ms})}$ are selected; delays are 0, 10, 20, 30 ms. The chemical effect starts after a 10.00000 ms delivery delay. Stable per-trial excitability offsets have Gaussian standard deviation 0.15000. These are chosen circuit quantities, not estimates of native CRY effects.

The simulator separately tracks go, cue detection and stop processes. Detection is available from the cue for 250.00000 ms. Stop accumulation begins after detection. Earliest eligible process ages are 350.00000, 40.00000 and 70.00000 ms for go, detection and stop. Temporal filtering uses 40.00000 ms and a 1.00000 s observation horizon. Stopping means a finite stop time earlier than the go time. Go means finite go time without earlier stopping; other trials are unresolved. The main article uses the temporally filtered weighted classifier. The majority, instantaneous weighted and mutually inhibiting competition rules operate on the same simulated releases and are supplied as sensitivity outputs rather than independent replicates. Synaptic stochasticity [9] and go-stop models [10, 11] motivate the framework but do not determine its parameters. Trigger failures remain a distinct issue [12].

The seeds are 26091711, 26091712 and 26091713, each with 8192 paired trials per condition. Let n_+ and n_- count positive and negative paired stopping discordances. The estimate is $100(n_+ - n_-)/n$ percentage points. For each count use a two-sided 97.50000% Clopper-Pearson interval with beta quantiles 0.01250 and 0.98750. If these intervals are $[l_+, u_+]$ and $[l_-, u_-]$, report $100[l_+ - u_-, u_+ - l_-]$. Bonferroni gives coverage at least 95% for each paired-difference interval under the simulated binomial marginals. These are per-condition intervals, not simultaneous coverage for the entire sweep, and contain no biological parameter uncertainty.

Case 1: 0.00000° , 1.00000 s, $f = 0.50000$; case 2: 1.00000° , 1.00000 s, $f = 0.50000$; case 3: 5.00000° , 1.00000 s, $f = 0.50000$; case 4: 1.00000° , 0.01000 s, $f = 0.50000$; case 5: 1.00000° , 1.00000 s, $f = 0.01000$; case 6: free rotation/equal input. For case

Table S4 Selected temporal readouts, in the order plotted in main-text Fig. 4. Each row has $n = 24576$. Counts allow interval reproduction without rerunning the stochastic simulator.

Case	Change (points)	Interval (points)	n_+	n_-
1	5.50944	[5.17006, 5.84444]	1354	0
2	3.38135	[3.11008, 3.64891]	831	0
3	0.05697	[0.01057, 0.10164]	14	0
4	0.00814	[-0.01715, 0.03305]	2	0
5	0.06917	[0.01929, 0.11722]	17	0
6	0.00000	[-0.01783, 0.01783]	0	0

6, $q_1 = q_0$ is explicitly imposed because the molecular difference is unresolved. This produces identical paired outcomes; it does not prove an exactly zero physical contrast. Original trials used NumPy 2.5.0. Regenerating figures and binomial intervals with another runtime does not regenerate those random streams. The bounded archive contains the full summary counts; raw event arrays remain in the local investigation and are excluded from this compact package.

For a single independent binary preparation, a maximal coupling makes corresponding units disagree with probability $|\Delta q|$. The chance that any of N units differs is $1 - (1 - |\Delta q|)^N$. Any identical downstream stochastic kernel cannot increase total variation [13]; hence the main-text probability bound. This derivation assumes one preparation, independent units and a common downstream kernel. It does not bound arbitrarily repeated writing or correlated sources using the same formula.

S7 Biological interpretation

The circuit receives molecular yields through an imposed conversion. Chemical delivery, channel occupancy and release probability are not calibrated as successive stages of a native pathway.

Several sources support individual candidate biological links while leaving the central bridge unmeasured. Fogle et al. [14], neuronal light-response experiments, and Fogle et al. [15], Hk rescue experiments, establish CRY-dependent light responses in their fly preparations. Baik et al. [16], Results, separate acute depolarisation from clock resetting. Pan et al. [17], Results, address cofactor oxidation and current in a mammalian Kv1.1–Kv β 1 preparation. Arthaut et al. [18], Results, address light-driven ROS production. Rorsman et al. [19], redox-input and current-inactivation experiments, concern a channel-associated chemical memory. Combining these observations motivates a route; it does not supply native dark-spin preparation, local chemical delivery, channel occupancy kinetics or synaptic gain. No tissue-level or cell-level effect size follows from the synthetic circuit.

Wong et al. [5], Section II and equation (8), support joint population inheritance; Section IV and Figures 4–6 challenge large enhancements under realistic retention and interaction choices. Their observable is directional anisotropy, not our history contrast. Kattnig et al. [4] support the relevance of fluctuating interactions, not the closed-shell

nuclear lifetime assumed by a fixed-orientation model. Neither reference is presented as an independent reproduction of the new calculations.

The operational stopping variable does not represent conscious intention. Accumulator-based accounts of readiness-potential averages [20] and criticism of their unique interpretation [21] provide competing contexts. Neither identifies a cryptochrome contribution. A molecular memory, a redox memory, a circuit integration time and a conscious-veto claim must remain separate until an experimental observation model connects them.

S8 Reproduction and verification scope

Online Resource 2 contains compact CSV/JSON result records, selected electronic-response arrays, the frozen reference NPZ, hashes, calculation source for inspection and `reproduce.py`. Running that script with Python, NumPy, SciPy, pandas and Matplotlib regenerates all four main figures and the numerical table fragments. It verifies input hashes, full-reset arithmetic, the nuclear/motion decomposition and neural intervals from discordance counts. SciPy supplies the statistical and numerical routines [22]. These checks are independent arithmetic re-evaluations of frozen results, not new physical simulations.

The archive is deliberately bounded: no complete upstream MD trajectories, SCF checkpoints or neural event arrays are included. The original calculation drivers and dependency source are supplied for inspection, with exact paths and missing-input limitations in the README. Reproducing electronic tensors from first principles requires the corresponding omitted structures/checkpoints and environments. No claim of complete end-to-end reproducibility is made by a successful figure rebuild.

Machine precision, angular truncation, stochastic sampling, reduced-motion approximation and physical parameter uncertainty are distinct. The public source repository contains the calculation code and compiled article and supplement. Results and large data, including Online Resource 2, are excluded; no public data deposit or persistent identifier is claimed.

References

- [1] Haberkorn R: Density matrix description of spin-selective radical pair reactions *Molecular Physics* **32** 1491–1493 (1976) <https://doi.org/10.1080/00268977600102851>
- [2] Fay T P, Lindoy L P and Manolopoulos D E: Spin-selective electron transfer reactions of radical pairs: Beyond the Haberkorn master equation *The Journal of Chemical Physics* **149** 064107 (2018) <https://doi.org/10.1063/1.5041520>
- [3] Lindblad G: On the generators of quantum dynamical semigroups *Communications in Mathematical Physics* **48** 119–130 (1976) <https://doi.org/10.1007/BF01608499>

- [4] Kattnig D R, Solov'yov I A and Hore P J: Electron spin relaxation in cryptochrome-based magnetoreception *Physical Chemistry Chemical Physics* **18** 12443–12456 (2016) <https://doi.org/10.1039/C5CP06731F>
- [5] Wong S Y, Solov'yov I A, Hore P J and Kattnig D R: Nuclear polarization effects in cryptochrome-based magnetoreception *J. Chem. Phys.* **154** 035102 (2021) <https://doi.org/10.1063/5.0038947>
- [6] Bisht K, Lomholt M A and Khandelia H: Sensing membrane voltage by reorientation of dipolar transmembrane peptides. *Biophysical Journal* **123**, 584–597 (2024). <https://doi.org/10.1016/j.bpj.2024.01.037>
- [7] Veinberg S L *et al.*: Practical considerations for the acquisition of ultra-wideline ^{14}N NMR spectra. *Solid State Nuclear Magnetic Resonance* **84**, 45–58 (2017). <https://doi.org/10.1016/j.ssnmr.2016.12.008>
- [8] Sun Q *et al.*: Recent developments in the PySCF program package. *The Journal of Chemical Physics* **153**, 024109 (2020). <https://doi.org/10.1063/5.0006074>
- [9] Branco T and Staras K: The probability of neurotransmitter release: variability and feedback control at single synapses *Nature Reviews Neuroscience* **10** 373–383 (2009) <https://doi.org/10.1038/nrn2634>
- [10] Logan G D and Cowan W B: On the ability to inhibit thought and action: A theory of an act of control. *Psychological Review* **91** 295–327 (1984) <https://doi.org/10.1037/0033-295X.91.3.295>
- [11] Boucher L, Palmeri T J, Logan G D and Schall J D: Inhibitory control in mind and brain: An interactive race model of countermanding saccades. *Psychological Review* **114** 376–397 (2007) <https://doi.org/10.1037/0033-295X.114.2.376>
- [12] Verbruggen F *et al.*: A consensus guide to capturing the ability to inhibit actions and impulsive behaviors in the stop-signal task *eLife* **8** e46323 (2019) <https://doi.org/10.7554/eLife.46323>
- [13] Gibbs A L and Su F E: On Choosing and Bounding Probability Metrics *International Statistical Review* **70** 419–435 (2002) <https://doi.org/10.1111/j.1751-5823.2002.tb00178.x>
- [14] Fogle K J, Parson K G, Dahm N A and Holmes T C: CRYPTOCHROME is a blue-light sensor that regulates neuronal firing rate *Science* **331** 1409–13 (2011) <https://doi.org/10.1126/science.1199702>
- [15] Fogle K J *et al.*: CRYPTOCHROME-mediated phototransduction by modulation of the potassium ion channel β -subunit redox sensor *Proc. Natl Acad. Sci. USA* **112** 2245–50 (2015) <https://doi.org/10.1073/pnas.1416586112>

- [16] Baik L S, Au D D, Nave C, Foden A J, Enriquez-Villalva W K and Holmes T C: Distinct mechanisms of *Drosophila* CRYPTOCHROME-mediated light-evoked membrane depolarization and in vivo clock resetting *Proceedings of the National Academy of Sciences* **116** 23339–23344 (2019) <https://doi.org/10.1073/pnas.1905023116>
- [17] Pan Y, Weng J, Cao Y, Bhosle R C and Zhou M: Functional Coupling between the Kv1.1 Channel and Aldoketoreductase Kv β 1 *Journal of Biological Chemistry* **283** 8634–8642 (2008) <https://doi.org/10.1074/jbc.M709304200>
- [18] Arthaut L D *et al.*: Blue-light induced accumulation of reactive oxygen species is a consequence of the *Drosophila* cryptochrome photocycle *PLoS One* **12** e0171836 (2017) <https://doi.org/10.1371/journal.pone.0171836>
- [19] Rorsman H O, Müller M A, Liu P Z, Sanchez L G, Kempf A, Gerbig S, Spengler B and Miesenböck G: Sleep pressure accumulates in a voltage-gated lipid peroxidation memory *Nature* **641** 232–9 (2025) <https://doi.org/10.1038/s41586-025-08734-4>
- [20] Schurger A, Sitt J D and Dehaene S: An accumulator model for spontaneous neural activity prior to self-initiated movement *Proc. Natl Acad. Sci. USA* **109** E2904–13 (2012) <https://doi.org/10.1073/pnas.1210467109>
- [21] Bogler C, Grujić B and Haynes J D: Clarifying the nature of stochastic fluctuations and accumulation processes in spontaneous movements *Front. Psychol.* **14** 1271180 (2023) <https://doi.org/10.3389/fpsyg.2023.1271180>
- [22] Virtanen P *et al.*: SciPy 1.0: fundamental algorithms for scientific computing in Python *Nature Methods* **17** 261–272 (2020) <https://doi.org/10.1038/s41592-019-0686-2>